Replication of Young-eun Lee & Susan A. Gelman (2022, Journal of Experimental Child Psychology)

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Table of contents

Introduction . . . 1
Methods . . . 3
- Power Analysis . . . 3
- Planned Sample . . . 3
- Materials . . . 3
- Procedure . . . 4
- Analysis Plan . . . 4
- Differences from Original Study . . . 5
- Methods Addendum (Post Data Collection) . . . 6

Results . . . 6
- Data preparation . . . 6
- Confirmatory analysis . . . 9
- Exploratory analyses . . . 11

Discussion . . . 16
- Summary of Replication Attempt . . . 16
- Commentary . . . 16

Introduction

The experiment I have chosen was conducted by Lee and Gelman (2022) to investigate whether, for children and adults, behaviors that are seen as violations of property rights when referring to physical property are similarly viewed as violations when referring to digital property. Since data will be collected via SONA, I am proposing replicating only the adult sample from this study. In this experiment, adults and children rated how okay/not okay the actions of a

character were after hearing stories about a person who downloaded a digital file that did not belong to them versus a person who took a physical item that did not belong to them. I am focusing on the result that adults found downloading a digital item to be more acceptable than taking a physical item (children's pattern of evaluations looked very similar to adults'); the authors tested this finding using a linear mixed model. They found a significant main effect of format type, LRT, $\chi^2(1) = 194.2$, b= -1.46, SE = 0.09, $p < .001$. The authors also conducted one-sample t tests (two-tailed) to test whether participants' ratings differed significantly from a neutral valence (0); I will also replicate this test with my data. They found that adults rated taking a physical item negatively when compared to neutral, $t(179) = -19.74$, $p < .001$, and rated taking a digital item neutrally, $t(179) = 1.62$, $p > .10$. I have run their code and successfully reproduce these numbers using their data.

Additionally, since SONA studies must be 30 minutes, I am also going to ask participants an open ended question justifying their choice. This was included in the original study ("What made you think so? Describe your reason below."), and the coding scheme the authors used is included in the paper.

This experiment is relevant to my own research interests as I have recently been interested in the development of children's understanding of digital technology - specifically, I have been thinking about children's thinking about their own digital privacy/ownership of their digital image in regard to their parents/caregivers sharing recordings of them. Understanding how people understand digital technology as similar or different to non-digital examples is relevant to this line of research. To this end, I may also try to increase my study length by adding an exploratory question(s) probing participants' attitudes towards the sharing of their digital image via video recordings (maybe thinking of their own parents).

All data and protocols are available at https://osf.io/3g2j4/, including example stimuli. Adults in this study were presented with six scenarios, randomly presented via Qualtrics, in which a target character either placed a physical item left next to a shared computer into their bag, or downloaded a file from a shared computer onto their own. Participants responded how okay or not okay the action was using a 5-point Likert scale. Three of the scenarios adults read were about physical objects and three were about digital objects (music, movie, book, photo of nature, drawing, and game; the types of items shown in a digital versus physical format were counterbalanced).

Since the stimuli used in this experiment were written vignettes presented via Qualtrics, I am wondering now if this procedure will still lend itself well to coding a javascript experiment, since that is something I would like to gain experience with. I also see a potential challenge with replicating the stimuli and procedures exactly, since only one example of stimuli is posted on OSF and the authors counterbalanced the presentation of certain vignettes in a way that is slightly opaque to me. I anticipate that I may have to reach out to the authors for their full procedure and materials, or recreate them myself. Another challenge I foresee is that my sample may differ from the original study's sample. The original sample consisted of adult participants recruited via Amazon MTurk in 2019. It is likely that the attitudes of

undergraduates in 2026 may differ from those of general adults in 2019, particularly when it comes to anything related to digital technology.

Learning goals: By doing this project, I hope to learn how to run an experiment with human participants on SONA, as well as coding a javascript experiment. This project will also help me acquire skills in data analysis, specifically wrangling/cleaning data in R and running a linear mixed model (which I have pretty limited experience with).

Link to repository: https://github.com/mikkahoffman/lee2022.git Link to paper: https://github.com/mikkahoffman/lee2022/blob/b99ec0bc5b82c91a985fb262e186f7811e3d0c25/original_paper/Lee%26Gelman_2022.pdf

Methods

Power Analysis

At the instructors' recommendation, I calculated the original effect size for the one-sampled t-test comparing adult non-digital ratings to a neutral valence of 0. The calculated effect size was d = -1.64 (95% CI [-2.03, -1.25]). Using this effect size, I calculated that original experiment had 100% power. I thus decided to use the original sample size of N=60 for my replication.

```
library(pwr)
pwr.t.test(d = -1.64, n = 60, type = "one.sample")$power
```

```
[1] 1
```

Planned Sample

The planned sample size is 60 adult participants. As the sample will be collected via SONA, participants will be predominantly undergraduate students, age 18-22.

Materials

"Adults read six scenarios in total (music, movie, book, photo of nature, drawing, and game) in orders randomized by Qualtrics, with three items presented in a digital format and the other three items presented in a physical format. In addition, the types of items shown in a digital versus physical format were counterbalanced." For example, for digital format scenarios, participants will read, "[X] finds that a book was saved on a shared computer in the room. [X] is the only person in the room. No one will know if [X] downloads the book on her computer. Is it okay or not okay for [X] to download the book on her computer? How okay or not

okay is it?" In the physical version, the story is identical except that the item is shown as a physical object (e.g.,"[X] finds that a book was left next to a shared computer in the room. [X] is the only person in the room. No one will know if [X] puts the book in her bag. Is it okay or not okay for [X] to put the book in her bag? How okay or not okay is it?"). In the original study, stimuli were presented and randomized by Qualtrics; in this study, stimuli will be presented and randomized via a web-based JsPsych experiment. Exact vignette text will be copied directly from the author-provided study materials.

I will also add in two additional new vignettes that will be similarly counterbalanced for format and randomly presented after the original 6 vignettes (so participants in this study will see 8 vignettes in total). These new items will be a photograph of a child and a photograph of an adult (to probe ideas of digital privacy for children versus adults). These will be presented at the end of the procedure for all participants.

Procedure

After providing informed consent, participants will see six stories presented in random order, followed by an additional two stories (8 total). After reading each story, they will rate how okay or not okay the action was on a 5-point Likert scale ("from -2 being totally not okay to +2 being totally okay"). They will also answer an open-ended question explaining each of their answers ("What made you think so? Describe your reason below.").

The full experiment can be accessed at: https://mikkahoffman.github.io/lee2022/

Analysis Plan

Following the exclusion criteria from the original study, participants will be excluded if they do not complete all types of scenarios.

"Answers to open-ended questions from all 60 adult participants were also coded by two independent coders" - in this study, open-ended questions will likely be coded by one coder (me) due to resource constraints. "We predetermined the categories for coding explanations based on pilot testing with adults." Coding categories are specified in Table 1 of the original paper and will be used to code open-ended questions in this study.

Raw data will be cleaned to remove identifiable and irrelevant data and will yield a clean tidy data file for adult ratings with subject ID, item, format type, and response (coded -2 to 2) as columns, and a .csv for adult explanations with columns for subject ID, item, format type, coding category, answer text, and coding (1 or 0 to indicate if the response was coded as including that coding category) - subject IDs will be repeated in this data frame as each subject will appear once in each coding category.

Clarify key analysis of interest here The key analysis of interest is the likelihood ratio test probing the main effect of format type in the linear mixed model.

"All statistical analyses were conducted with R statistical software (R Version 4.1.1; R Core Team, 2021). We analyzed ratings with a linear mixed model (LMM) using the R package ' 'lme4" (Bates et al., 2015) and analyzed justifications with a generalized linear mixed model (GLMM) using the R package ' 'glmmTMB" (Brooks et al., 2017). Our full LMMs on ratings included format type, item type, children's age in months [not relevant for this study], and all interactions among the variables as fixed effects and participant identity (ID) as a random effect. Our full GLMMs on justification included format type, coding category, and the interaction between the variables as fixed effects and participant ID as a random effect. We first compared the full model with a null model that included only participant ID as a random intercept. If the full model provided a significantly better fit to the data than the null model, we then conducted hypothesis-driven tests to examine the role of individual predictors by sequentially dropping them from the full model and assessed changes in model fit using likelihood ratio tests (LRTs)."

To replicate the original analyses, these analyses will be run just on the original 6 vignettes.

Additional analyses: I will also conduct follow-up one-sample t-tests (two-tailed) "to examine whether ratings differed significantly from a neutral valence of 0." I also will analyze the justifications provided using the plan detailed above. I will also repeat the analyses detailed above with the full 8 vignettes to explore the effect of the two new vignettes.

Differences from Original Study

Differences in sample: The original sample was collected from adults ("M = 37.7 years, range = 24–66; 23 female and 37 male") on Amazon Mechanical Turk in 2019. This sample will be collected from undergraduate adults in 2026. Likely these samples will be different, as the age range of my sample will likely be narrower and younger, and there may be generational differences between these two samples. Their experiences with technology may be different, potentially leading to differences in their responses.

Differences in procedure: The original experiment was presented, randomized, and collected via Qualtrics. In this study, I will present, randomize, and collect data via a JsPsych experiment that I code. While I do not anticipate this creating meaningful differences in participants' experiences, there may be differences in things such as font size, scale presentation, etc. An additional difference is that participants will see two new vignettes following the original procedure. Furthermore, I have added in an attention check question and an AI honesty check that were not included in the original procedure,

Differences in analysis plan: As the original study was collected before generative AI tools were widely used, they did not account for screening out potentially AI generated answers in their data cleaning. By using their methods, I would similarly not be screening out potential AI answers - this could introduce poor-quality data into my sample. I have thus added in both an attention check question and an AI honesty question, and will exclude participants who do not pass either. I also will not be excluding participants on the basis of their IP

addresses not being based in the U.S., as this is not a huge concern when pulling from a UCSD undergraduate participant pool.

Methods Addendum (Post Data Collection)

You can comment this section out prior to final report with data collection.

Actual Sample

Sample size, demographics, data exclusions based on rules spelled out in analysis plan

Differences from pre-data collection methods plan

Any differences from what was described as the original plan, or "none".

Results

Data preparation

Data preparation following the analysis plan.

```
### Data Preparation

#### Load Relevant Libraries and Functions
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(data.table)
```

```
Attaching package: 'data.table'

The following objects are masked from 'package:lubridate':

    hour, isoweek, isoyear, mday, minute, month, quarter, second, wday,
    week, yday, year

The following objects are masked from 'package:dplyr':

    between, first, last

The following object is masked from 'package:purrr':

    transpose

The following object is masked from 'package:base':

    %notin%
```

```
library(lme4)
```

```
Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

    expand, pack, unpack
```

```
library(glmmTMB)
library(readr)
library(effectsize)

#### Import all data
#### get a vector of all csv file paths in data folder
 files <- list.files(path = "/Users/m5hoffman/Documents/GitHub/lee2022/data/raw/pilot_a_data
#### read and combine into one dataframe
 data <- read_csv(files)
```

```
Rows: 44 Columns: 15
```

```
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (11): stimulus, response, trial_type, plugin_version, subject_id, condit...
dbl  (4): rt, trial_index, time_elapsed, rating

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
 #maybe check how many participants were in each condition/responses for each item here and v
# hist(data$rating)
# table(data$item_id)
# table(data$condition)

# check for attention check, AI check answers, and participants who didn't rate all scenarios
# participant_exclusions_incomplete <- data |>
  # filter(is.na(item)) |>   #filter only for rating trials
  # filter(is.na(rating)) #identifies participants that did not complete a trial - these part
# participant_exclusions_checks <- data |> #identify participants that did not answer correct
#  filter(attention_check_response != 1) |>
#  filter(honesty_response != "no")


#### tidy up dataframe - select only relevant columns and responses
 data <- data |>
   select(subject_id, item, format_type, rating, justification) #select only relevant variabl

#### Data exclusion / filtering
#exclude participants who did not rate all scenarios
 participant_exclusions <- data |>
   filter(is.na(rating)) #identifies participants that did not complete a trial - these parti

#### Prepare data for analysis
# create one data frame for ratings with 6 original vignettes, and one with all 8 vignettes

data_replication <- data |>
  filter(item!="kidphoto",
         item!="adultphoto") |>
  select(-justification)

data_exploratory <- data |>
  select(-justification)
```

```
# create another data frame for justifications, including subject ID, format, item, category
# import data after hand coding
```

Confirmatory analysis

Copied from original paper's analyses:

```
doca.adult.full <- lmer(rating ~ format_type*item + (1|subject_id),
                        data = data_replication, REML=FALSE, na.action=na.omit) #full model t
```

```
fixed-effect model matrix is rank deficient so dropping 3 columns / coefficients
```

```
doca.adult.null <- lmer(rating ~ (1|subject_id), data = data_replication, REML=FALSE, na.acti
  anova(doca.adult.full, doca.adult.null) #likelihood ratio test
```

```
Data: data_replication
Models:
doca.adult.null: rating ~ (1 | subject_id)
doca.adult.full: rating ~ format_type * item + (1 | subject_id)
                npar    AIC     BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
doca.adult.null    3 82.557  86.091 -38.279    76.557
doca.adult.full   11 90.102 103.061 -34.051    68.102 8.4548  8     0.3903
```

```
  drop1(doca.adult.full, test="Chisq")
```

```
Single term deletions

Model:
rating ~ format_type * item + (1 | subject_id)
                 npar    AIC    LRT Pr(Chi)
<none>                90.102
format_type:item    2 89.733 3.6304  0.1628
```

```
#If the full model provided a significantly better fit to the data than the null model, we th

doca.adult.m1 <- update(doca.adult.full, ~ . -format_type:item)
  anova(doca.adult.full, doca.adult.m1)
```

```
Data: data_replication
Models:
doca.adult.m1: rating ~ format_type + item + (1 | subject_id)
doca.adult.full: rating ~ format_type * item + (1 | subject_id)
                npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
doca.adult.m1      9 89.733 100.33 -35.866    71.733
doca.adult.full   11 90.102 103.06 -34.051    68.102 3.6304  2     0.1628
```

```
  drop1(doca.adult.m1, test="Chisq")
```

```
Single term deletions

Model:
rating ~ format_type + item + (1 | subject_id)
            npar    AIC    LRT Pr(Chi)
<none>           89.733
format_type    1 90.598 2.8648 0.09053 .
item           5 84.017 4.2839 0.50931
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
doca.adult.m2 <- update(doca.adult.full, ~ . -format_type:item -item)
  anova(doca.adult.m2, doca.adult.m1)
```

```
Data: data_replication
Models:
doca.adult.m2: rating ~ format_type + (1 | subject_id)
doca.adult.m1: rating ~ format_type + item + (1 | subject_id)
              npar    AIC     BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
doca.adult.m2    4 84.017  88.729 -38.008    76.017
doca.adult.m1    9 89.733 100.335 -35.866    71.733 4.2839  5     0.5093
```

```
  drop1(doca.adult.m2, test="Chisq")
```

```
Single term deletions

Model:
rating ~ format_type + (1 | subject_id)
            npar    AIC     LRT Pr(Chi)
<none>           84.017
format_type    1 82.557 0.54057  0.4622
```

```r
summary(doca.adult.m2) ## final model
```

```
Linear mixed model fit by maximum likelihood  ['lmerMod']
Formula: rating ~ format_type + (1 | subject_id)
   Data: data_replication

      AIC       BIC    logLik -2*log(L)  df.resid
     84.0      88.7     -38.0      76.0        20

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.7114 -1.0456  0.1434  0.5256  1.9150

Random effects:
 Groups     Name        Variance Std.Dev.
 subject_id (Intercept) 0.2486   0.4986
 Residual               1.2167   1.1030
Number of obs: 24, groups:  subject_id, 4

Fixed effects:
                     Estimate Std. Error t value
(Intercept)           0.08333    0.40440   0.206
format_typenondigital -0.33333    0.45031  -0.740

Correlation of Fixed Effects:
            (Intr)
frmt_typnnd -0.557
```

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

```r
# do ratings differ from neutral?
doca.adult.digital <- subset(data_replication, format_type == "digital")
doca.adult.nondigital <- subset(data_replication, format_type == "nondigital")

t.test(doca.adult.digital$rating, mu=0)
```

```
    One Sample t-test

data:  doca.adult.digital$rating
t = 0.2664, df = 11, p-value = 0.7949
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 -0.6051689  0.7718356
sample estimates:
 mean of x
0.08333333
```

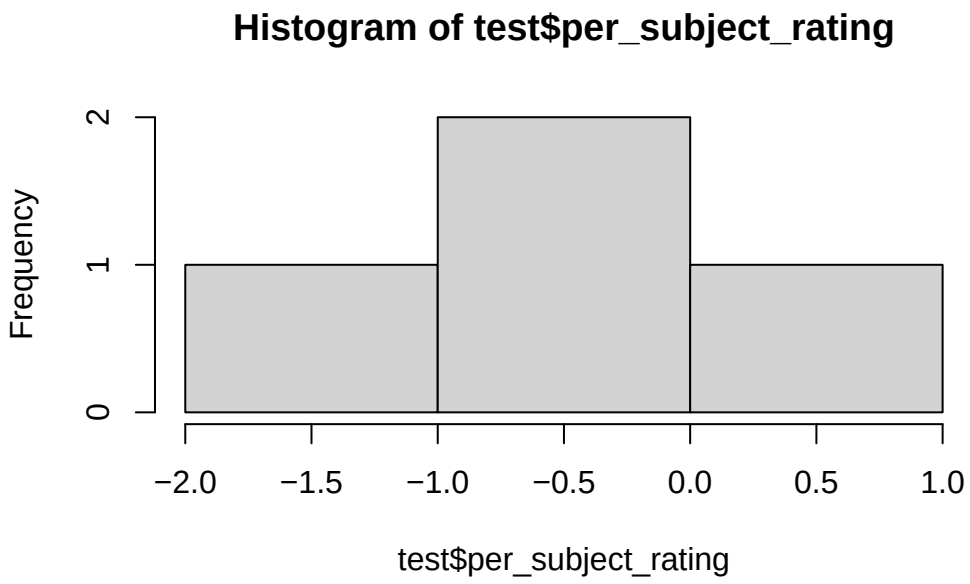
```r
t.test(doca.adult.nondigital$rating, mu=0)
```

```
    One Sample t-test

data:  doca.adult.nondigital$rating
t = -0.60892, df = 11, p-value = 0.5549
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 -1.1536394  0.6536394
sample estimates:
mean of x
    -0.25
```

```r
#effect size for non digital when summarizing per subject (replicating how I calculated effe
test <- doca.adult.nondigital %>%
  group_by(subject_id) %>%
  summarize(per_subject_rating = mean(rating))

hist(test$per_subject_rating)
```




```
t.test(test$per_subject_rating, mu=0)
```

```
    One Sample t-test

data:  test$per_subject_rating
t = -0.50233, df = 3, p-value = 0.65
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 -1.833839  1.333839
sample estimates:
mean of x
    -0.25
```

```
cohens_d(test$per_subject_rating, mu=0)
```

```
Cohen's d |        95% CI
-------------------------
-0.25     | [-1.23, 0.77]
```

```
# Adult: Do their explanations differ by format type?
#doca.adult.why.full <- glmmTMB(coding ~ category*format.type + (1|subjectid),
 #                                      family = binomial, data = doca.adult.why, na.action

#doca.adult.why.null <- glmmTMB(coding ~ (1|subjectid),
 #                                      family = binomial, data = doca.adult.why, na.action

  #  anova(doca.adult.why.full, doca.adult.why.null, test="LRT")
   # drop1(doca.adult.why.full, test="Chisq")

## Unpack the interaction between Category and Format Type
    ## ownership
  #  doca.adult.why.ownership <- subset(doca.adult.why, category == "Ownership")
   # doca.adult.why.ownership.full <- glmmTMB(coding ~ format.type + (1|subjectid),
  #                                          family = binomial, data = doca.adult.why.owners

  #  doca.adult.why.ownership.null <- glmmTMB(coding ~ (1|subjectid),
   #                                          family = binomial, data = doca.adult.why.owners

    #anova(doca.adult.why.ownership.full, doca.adult.why.ownership.null, test="LRT")
  #  summary(doca.adult.why.ownership.full)

    ## legality
   # doca.adult.why.legality <- subset(doca.adult.why, category == "Legality")
   # doca.adult.why.legality.full <- glmmTMB(coding ~ format.type + (1|subjectid),
   #                                        family = binomial, data = doca.adult.why.legalit

    #doca.adult.why.legality.null <- glmmTMB(coding ~ (1|subjectid),
     #                                       family = binomial, data = doca.adult.why.legalit

   #   anova(doca.adult.why.legality.full, doca.adult.why.legality.null, test="LRT")
    #  summary(doca.adult.why.legality.full)

    ## morality
  #  doca.adult.why.morality <- subset(doca.adult.why, category == "Morality")
   # doca.adult.why.morality.full <- glmmTMB(coding ~ format.type + (1|subjectid),
    #                                        family = binomial, data = doca.adult.why.moralit

    #doca.adult.why.morality.null <- glmmTMB(coding ~ (1|subjectid),
     #                                       family = binomial, data = doca.adult.why.moralit

      #anova(doca.adult.why.morality.full, doca.adult.why.morality.null, test="LRT")
```

```
    ## harmlessness
#    doca.adult.why.harm <- subset(doca.adult.why, category == "Harm")
#    doca.adult.why.harm.full <- glmmTMB(coding ~ format.type + (1|subjectid),
#                                        family = binomial, data = doca.adult.why.harm) # mod

    ## duplicability
#    doca.adult.why.duplicability <- subset(doca.adult.why, category == "Duplicability")
#    doca.adult.why.duplicability.full <- glmmTMB(coding ~ format.type + (1|subjectid),
#                                                 family = binomial, data = doca.adult.why.du

#    doca.adult.why.duplicability.null <- glmmTMB(coding ~ (1|subjectid),
#                                                 family = binomial, data = doca.adult.why.du

#      anova(doca.adult.why.duplicability.full, doca.adult.why.duplicability.null, test="LRT"
#      summary(doca.adult.why.duplicability.full)

    ## public nature
#    doca.adult.why.public <- subset(doca.adult.why, category == "Public")
#    doca.adult.why.public.full <- glmmTMB(coding ~ format.type + (1|subjectid),
#                                          family = binomial, data = doca.adult.why

#    doca.adult.why.public.null <- glmmTMB(coding ~ (1|subjectid),
#                                          family = binomial, data = doca.adult.why.public)

#      anova(doca.adult.why.public.full, doca.adult.why.public.null, test="LRT")
#      summary(doca.adult.why.public.full)

    ## item value
#    doca.adult.why.itemvalue <- subset(doca.adult.why, category == "ItemValue")
#    doca.adult.why.itemvalue.full <- glmmTMB(coding ~ format.type + (1|subjectid),
#                                             family = binomial, data = doca.adult.why.itemva

#    doca.adult.why.itemvalue.null <- glmmTMB(coding ~ (1|subjectid),
#                                             family = binomial, data = doca.adult.why.itemval

#      anova(doca.adult.why.itemvalue.full, doca.adult.why.itemvalue.null, test="LRT")

    ## possibility of being caught
#    doca.adult.why.caught <- subset(doca.adult.why, category == "Caught")
#    doca.adult.why.caught.full <- glmmTMB(coding ~ format.type + (1|subjectid),
#                                          family = binomial, data = doca.adult.why.caught)
```

```
#    doca.adult.why.caught.null <- glmmTMB(coding ~ (1|subjectid),
#                                          family = binomial, data = doca.adult.why.caught)

#      anova(doca.adult.why.caught.full, doca.adult.why.caught.null, test="LRT")

    ## conditional response
#    doca.adult.why.conditional <- subset(doca.adult.why, category == "Conditional")
#    doca.adult.why.conditional.full <- glmmTMB(coding ~ format.type + (1|subjectid),
#                                               family = binomial, data = doca.adult.why.cond

#    doca.adult.why.conditional.null <- glmmTMB(coding ~ (1|subjectid),
#                                               family = binomial, data = doca.adult.why.cond

#      anova(doca.adult.why.conditional.full, doca.adult.why.conditional.null, test="LRT")
#      summary(doca.adult.why.conditional.full)
```

Discussion

Summary of Replication Attempt

Open the discussion section with a paragraph summarizing the primary result from the confirmatory analysis and the assessment of whether it replicated, partially replicated, or failed to replicate the original result.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis, (b) assessment of the meaning of the replication (or not) - e.g., for a failure to replicate, are the differences between original and present study ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the replication attempt. None of these need to be long.